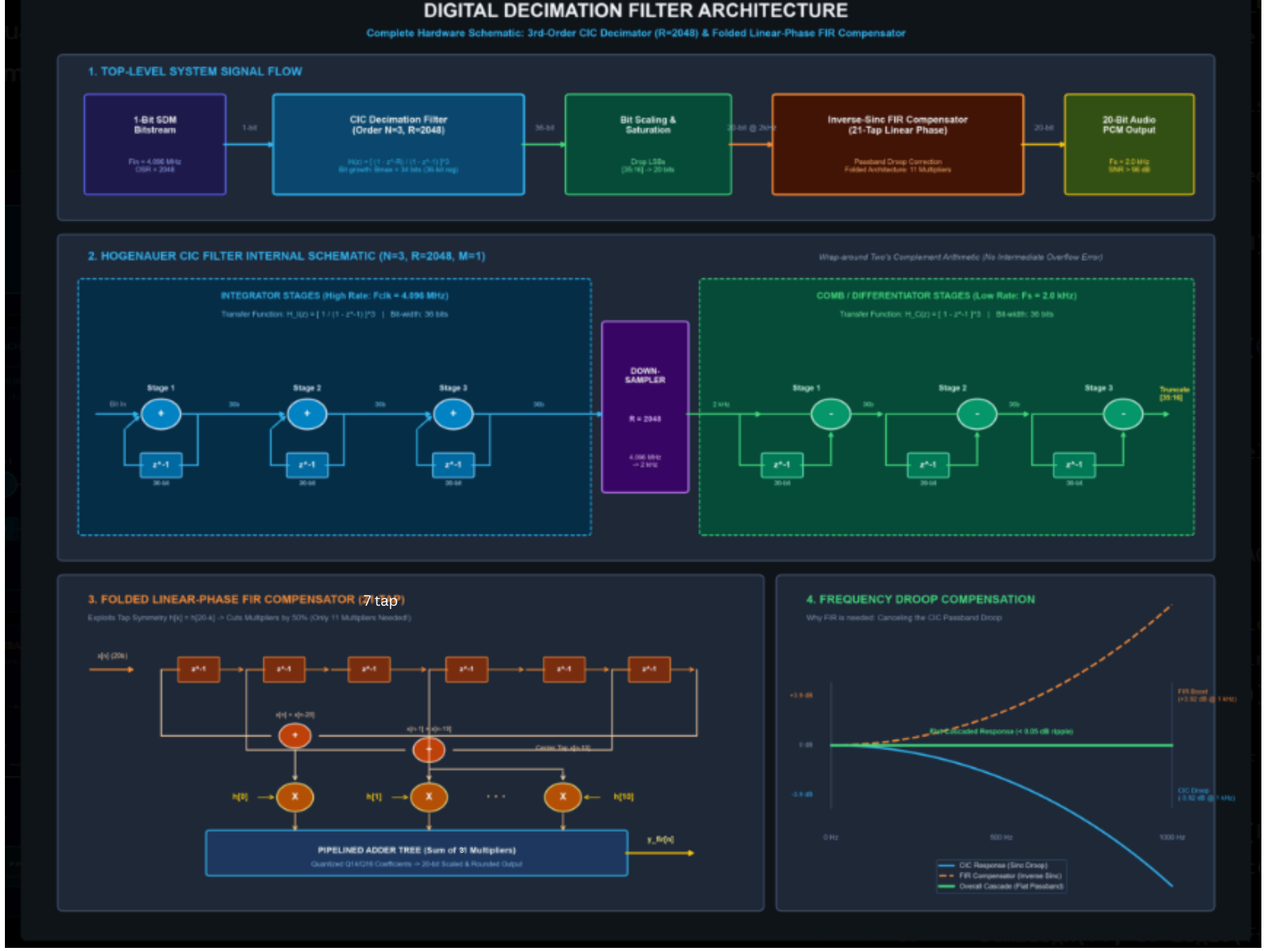
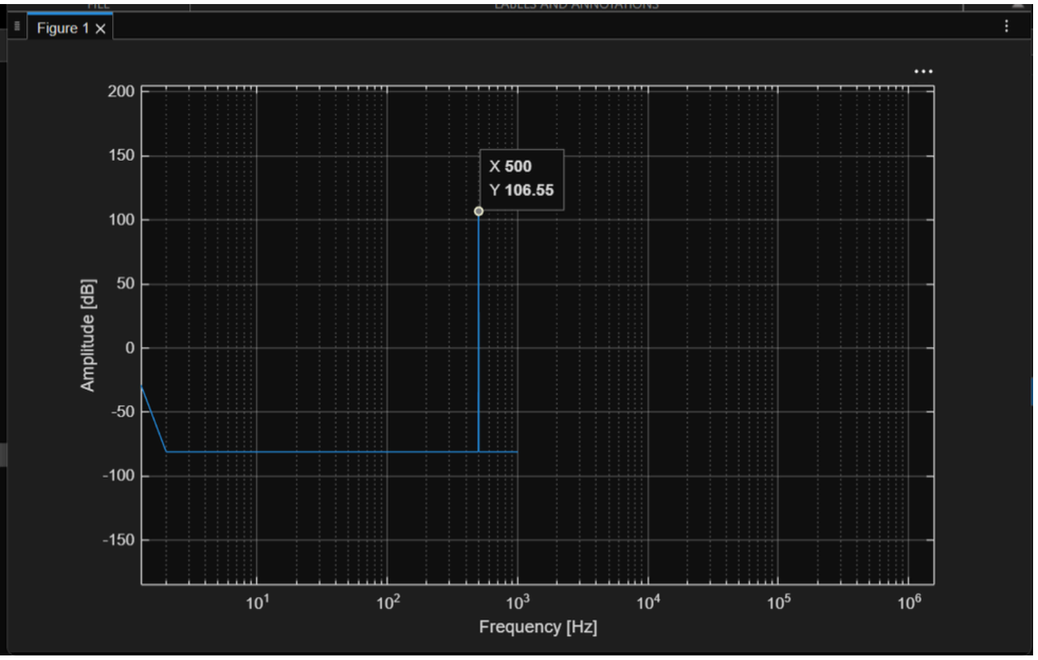



```
  FIR FILTER COEFFICIENT DESIGN (L = 7 Taps, 4 Multipliers)
================================================================
Center Tap: h(3) = 1.626979  -->  Q14 Integer: 26656
Tap h(0) = h(6) = -0.090063  -->  Q14 Integer:  -1476
Tap h(1) = h(5) =  0.175991  -->  Q14 Integer:   2883
Tap h(2) = h(4) = -0.431911  -->  Q14 Integer:  -7076

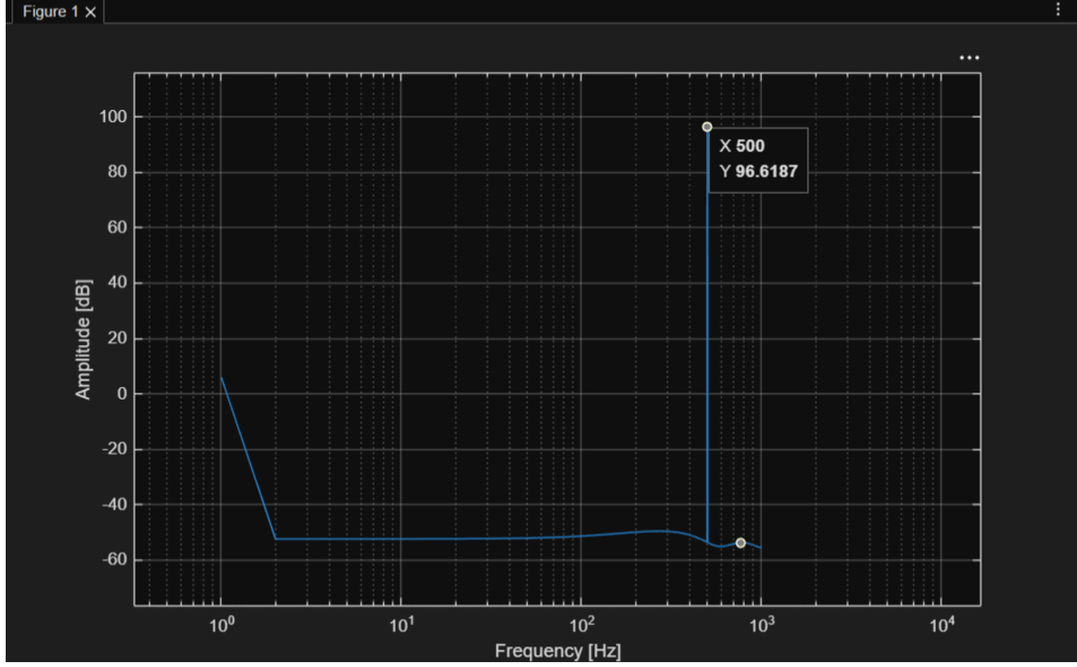
----------------------------------------------------------------
Verilog localparam Statements for fir.v:
----------------------------------------------------------------
localparam signed [COEFF_WIDTH-1:0] H0 = 16'sd-1476;
localparam signed [COEFF_WIDTH-1:0] H1 = 16'sd2883;
localparam signed [COEFF_WIDTH-1:0] H2 = 16'sd-7076;
localparam signed [COEFF_WIDTH-1:0] H3 = 16'sd26656; // Center tap
================================================================
```

- I have used Least mean square method to derive the coefficients of Fir filter because we need strong brick low passfilter , and it avoids ripple error, and it has a good snr
- to reduce latency i have taken length of fir to be 7 otherwise 11 would be optimal

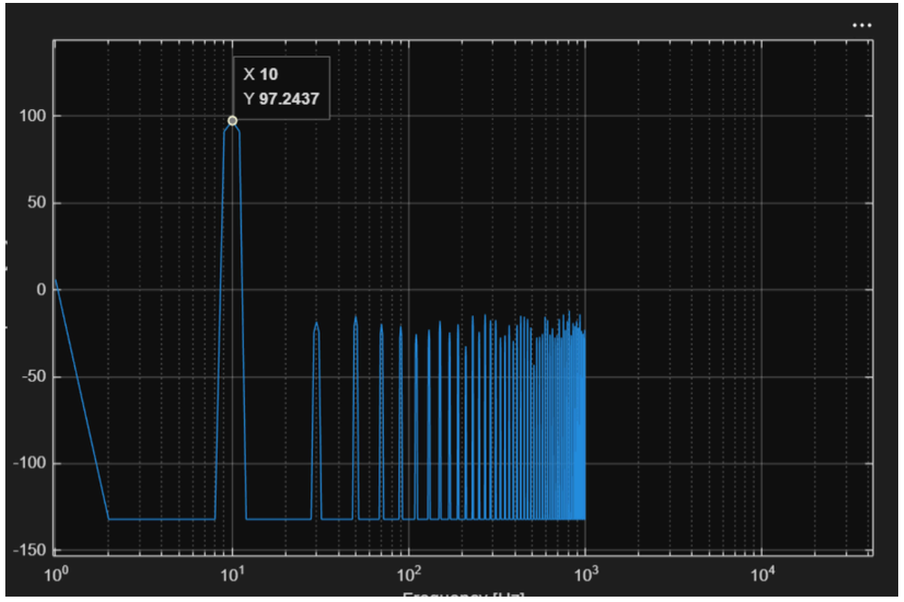
N=3 because there is not much significant improvement of snr by increasing N further



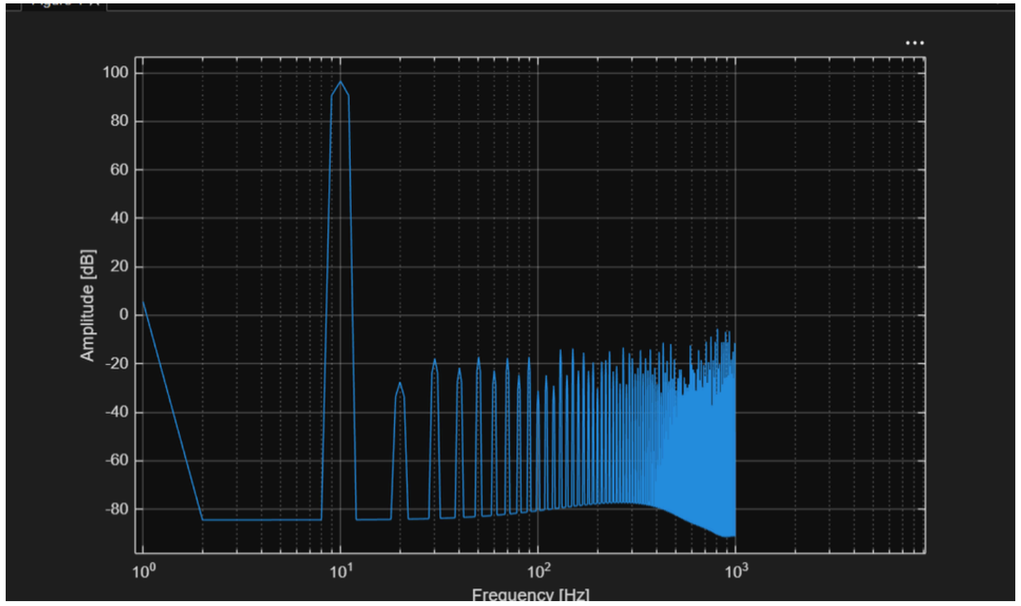
CIC OUTPUT OF 500Hz



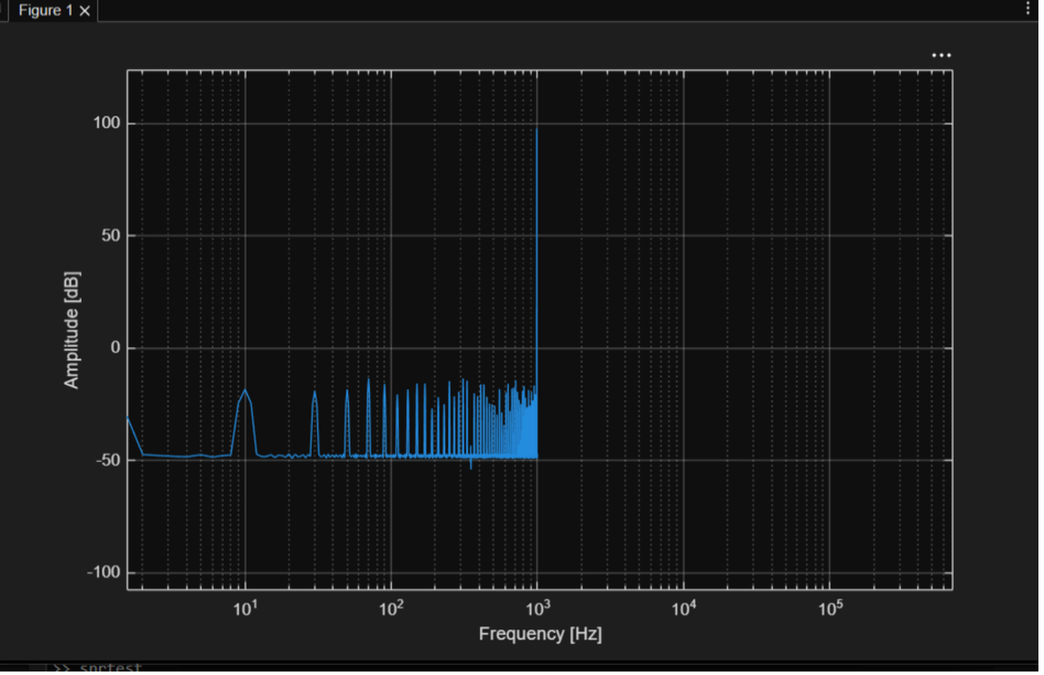
FINAL OUTPUT OF 500Hz



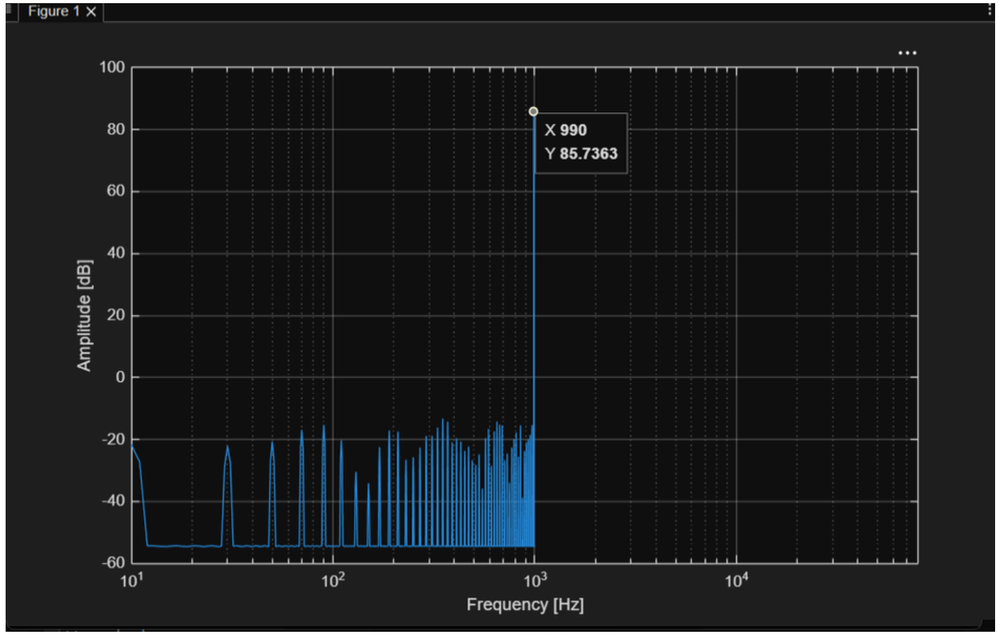
CIC OUTPUT 0F 10Hz



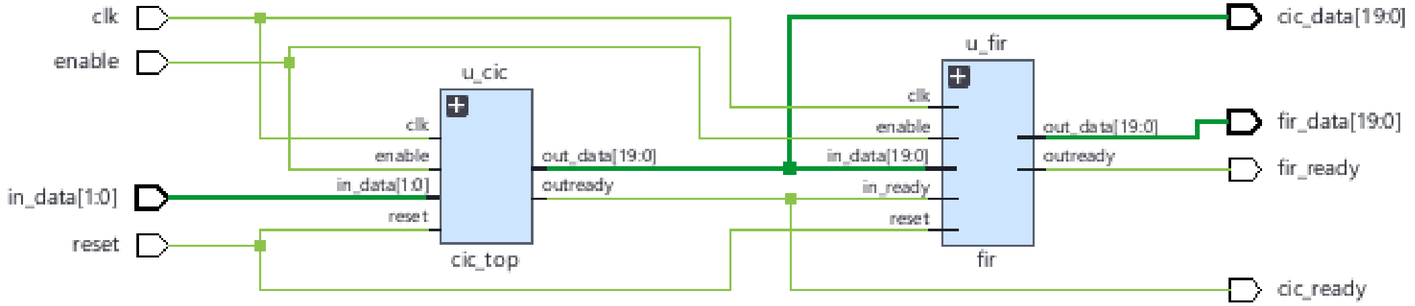
FIR OUTPUT OF 10Hz



fir output of 990hz



fir output of 990 Hz


clk
enable
in_data[1:0]
reset
u_cic
clk
enable
in_data[1:0]
reset
out_data[19:0]
outready
cic_top
u_fir
clk
enable
in_data[19:0]
in_ready
reset
out_data[19:0]
outready
fir
cic_data[19:0]
fir_data[19:0]
fir_ready
cic_ready

